

AJAY JANGID

Assistant Vice President, AI & GenAI Engineering | Staff AI / Lead GenAI Engineer

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PROFESSIONAL SUMMARY

AVP-level AI Engineer with 7+ years designing and shipping production GenAI systems at scale. Deep hands-on expertise in RAG pipelines, LLM fine-tuning (SFT/RLHF), agentic workflow automation (CrewAI/LangChain), self-hosted ASR and inference infrastructure (vLLM, air-gapped GPU serving), and enterprise MLOps/LLMOps. Track record of leading cross-functional engineering efforts in regulated fintech/banking environments — from proof-of-concept to 99.9% uptime production systems — while mentoring teams, running model evaluation programs, and defining technical roadmaps. Passionate about building AI systems that are measurably better: faster inference, higher recall, lower hallucination rate.

CORE TECHNICAL COMPETENCIES

GenAI & LLMs: RAG (dense + hybrid retrieval), LLM fine-tuning (SFT, LoRA, RLHF), prompt engineering, PEFT, instruction tuning

Agentic Systems: Multi-agent orchestration (CrewAI, LangChain Agents), LlamaIndex, tool-use pipelines, ReAct / Plan-and-Execute patterns, text-to-SQL and workflow-automation agents

Inference Infrastructure: vLLM (OpenAI-compatible serving, KV-cache/prefix-cache optimization), TGI, Triton Inference Server, quantization (GPTQ, AWQ, GGUF), multi-GPU replica architectures, GPU batch serving, latency optimization

Speech / ASR: Whisper (fine-tuning, CTranslate2 conversion), IndicConformer, batch + near-real-time ASR, streaming ASR, speaker diarization, telephony (8kHz G.711) audio processing

Vector & Retrieval: FAISS, Pinecone, Weaviate, Chroma, AWS Kendra, Elasticsearch hybrid search, reranking (Cohere, cross-encoder), knowledge graphs, RAG guardrails & hallucination control

Models & Frameworks: GPT-4, LLaMA 3, Mistral, Whisper, BERT, RoBERTa, Transformers (HuggingFace), PyTorch

MLOps & LLMOps: AWS SageMaker, Azure ML, GCP, Cloudera CML, MLflow, Docker, Kubernetes, CI/CD, prompt versioning, evaluation & observability pipelines (LLM traces, latency/accuracy dashboards)

Data & Streaming: PySpark, Spark, Apache Kafka, PostgreSQL, MongoDB, MySQL

Languages & APIs: Python (expert), SQL, FastAPI, Flask, LangChain, LlamaIndex

Leadership: Technical roadmap ownership, cross-functional team leadership, mentoring (interns + engineers), model evaluation frameworks, AI governance & PII/compliance (KYC, AML)

PROFESSIONAL EXPERIENCE

Axis Bank *Mumbai, India Jun 2023 – Present*

Assistant Vice President — AI & GenAI Engineering *Apr 2026 – Present*

Senior Data Science Manager — AI & GenAI Engineering *Jun 2023 – Mar 2026*

Leading end-to-end development of enterprise GenAI platforms in banking — Voice AI, agentic workflows, and enterprise automation — integrating LLMs into core banking processes to cut operational cost, turnaround time, and vendor reliance.

- Built and lead the bank's core AI Engineering and GenAI platform — an enterprise-wide **AI-Powered Call Analytics Intelligence Platform** processing **1.2 million+ customer interactions monthly** (2,000 call hours/day, 60,000+ AI inference requests/day) for **18,000+ contact center agents**, delivering automated intent detection, sentiment analysis, compliance monitoring, quality scoring, and agent performance analytics.
- Designed and owned a production-grade **Retrieval-Augmented Generation (RAG) platform** leveraging GPT-4, Gemini, Llama, and Whisper to automate analysis of **1M+ customer calls monthly**, achieving **80%+ extraction accuracy** while reducing manual audit effort and generating **₹2 crore+ annual operational savings**.
- Architected **agentic AI workflows using CrewAI/LangChain** to automate core banking operations: **text-to-SQL agents** for analytics automation (SQL validation, schema grounding, query-consistency checks), **email automation agents** for triage, summarization, and response drafting, and **Risk & Compliance AI agents** for document intelligence, regulatory mapping, exception handling, and audit trails.
- Architected a self-hosted, air-gapped LLM/ASR serving layer on Cloudera CML (AWS G5 GPUs) built around **vLLM OpenAI-compatible endpoints** and a **TP1×4 independent-replica GPU architecture** achieving **80–90% KV-cache prefix-hit rates**, plus batch and near-real-time **ASR** (benchmarked/fine-tuned IndicConformer and Whisper large-v3, selected ARTPARK-II Sc SraVaani-1.0) for speech recognition, diarization, and dialog orchestration; built the AVP-level business case — GPU reservation economics, cost-per-call modeling, throughput projections — to migrate off a legacy batch Ray Serve pipeline.

- Built a high-performance **Entity & Information Extraction Engine** using fine-tuned transformer models (LoRA-adapted BERT variants), achieving **85%+ F1-score** and replacing an incumbent vendor solution while reducing annual operating costs from **₹1.2 crore+ to approximately ₹30 lakh**.
- Built model evaluation and RAG-safety infrastructure — a **Cohen's κ inter-rater harness** for a 20-parameter call-scoring rubric, and **guardrails, hallucination control, and eval/safety gates** across the retrieval stack (Pinecone, Weaviate, Chroma, AWS Kendra, knowledge graphs, hybrid search) — plus relationship-manager performance analytics (WoW/MoM trends) across roughly **60,000 calls/day and 10,000–12,000 RMs**.
- Owns the platform's **MLOps + LLMops lifecycle** — CI/CD, prompt versioning, evaluation metrics, and observability — with secure deployment on Azure/AWS + Kubernetes, PII masking, role-based access, and AI governance aligned to KYC/AML, maintaining **zero regulatory observations** and **99.9% production SLA** across mission-critical AI services.
- Drove in-house **GenAI capability building for marketing** — an AI-driven Customer Intelligence & Marketing Personalization Platform (GPT-4, Gemini) automating campaign generation and personalization — driving **15% improvement in campaign engagement** and **₹50 lakh+ annual savings** while reducing external vendor spend.
- Mentored and led a multidisciplinary AI team comprising Data Scientists and ML Engineers — including intern-led builds such as a NetworkX knowledge-graph platform for earnings-call and analyst intelligence, and a customer journey insights platform — establishing engineering standards, hiring processes, and AI platform best practices.

Accenture AI consulting *India · Remote May 2022 – Jun 2023*

Senior Data Scientist

- Developed and deployed a graph-based anomaly detection solution for unstructured data, improving detection accuracy by **25%** and reducing false positives by **30%**.
- Engineered advanced search and ranking algorithms for identifying emerging risks in the insurance claims domain, using transformer models (BERT, RoBERTa) for question-answering tasks.
- Developed a dynamic phrase-ranking framework combining contextual word embeddings with time-series analysis of historical and real-time data trends to improve model performance.
- Optimized risk identification and ranking pipelines at scale, improving predictive accuracy by **20%** and significantly reducing false positives in risk assessment.
- Streamlined ML integration into enterprise systems, enhancing operational efficiency by **20%** while processing millions of records daily, and reduced operational risk by **15%** through actionable, data-driven insights.

BRIDGEi2i Analytics Solutions *India · Remote Jan 2021 – May 2022*

Data Science – Analytics Consultant

- Architected and deployed a high-performance extraction engine, reducing insurance claim processing time from 1–2 days to 5–10 minutes — over **90% efficiency improvement**.
- Developed a scalable OCR-integrated pipeline processing **10,000+ PDF documents monthly**, paired with NLP models achieving **98% entity extraction accuracy**.
- Designed a decision engine leveraging complex business rules to automate claim validation (accepted / rejected / review), delivering a **95% reduction in manual errors** and a **40% increase in processing throughput**.

Code Matrix Software India Pvt Ltd *Remote Feb 2020 – Dec 2020*

Machine Learning Engineer

- Built a production hybrid recommendation system (semantic vector search + collaborative filtering via ElasticSearch), scaling to **10M+ queries/day** with sub-second response times and **99.9% uptime** on AWS.
- Increased search relevance by **20%**, and boosted personalized recommendation click-through rates by **15%** and conversion rates by **10%** through real-time user-behavior analysis.
- Scaled the system to handle a **30% increase in traffic** with minimal latency impact (<200ms), while reducing computational costs by **25%** through inference and infrastructure optimization.

Analysed.in *India · Remote May 2019 – Feb 2020*

Machine Learning Engineer

Built an AI-powered virtual interview platform — an early predecessor to modern conversational AI interview systems.

- Developed NLP models for resume parsing, skill extraction, and candidate-job matching, and ML models for candidate scoring and recommendation; built REST APIs (Flask) to serve models in production.
- Improved model accuracy through feature engineering, hyperparameter tuning, and cross-validation (TensorFlow, Keras, Scikit-learn); built dashboards to analyze interview performance and candidate insights.

STYLABS *Mumbai, India Jul 2018 – Feb 2019*

Machine Learning Engineer

- Developed Generative Adversarial Network (GAN) models for image synthesis and computer vision solutions for real-time image/video analytics.
- Built object detection pipelines using YOLO v2 for surveillance/video-processing applications, with OpenCV-based preprocessing, augmentation, and feature-tracking; trained and evaluated CNN models (TensorFlow/Keras) via transfer learning, assessed on precision, recall, and mAP.

PUBLICATIONS, WRITING & PROJECTS

Contributing Writer, Medium (Apr 2019 – Present) — writes on applied AI/ML and GenAI engineering topics.

Author, [Agentic AI Architect — Interactive Learning Portal](#) — self-published, 23-chapter interactive curriculum on designing and architecting production agentic AI systems.

Author, [The RAG Architect's Handbook](#) — self-published, 27-lesson interactive guide to Retrieval-Augmented Generation, from first principles to production architecture.

EDUCATION

M.Tech, Machine Learning & Artificial Intelligence 2022 – 2024

BITS Pilani, India

B.Tech, Electronics & Telecommunication 2014 – 2018

Shree L.R. Tiwari College of Engineering, Mumbai

AWARDS & RECOGNITION

- First Runner-Up — Accenture Data Science Hackathon (50+ teams competing nationally)
- Spot Award — High-impact AI delivery, Accenture
- Quarter Team Award — Excellence in AI innovation, Bridgei2i

CERTIFICATIONS

- Applied AI — Masters in Applied Machine Learning
- Text Retrieval & Search Engines — Coursera (University of Illinois)